

CREDIT APPROVAL SYSTEM - DETAILED PROJECT SUMMARY

1. PROJECT COMPLETION STATUS

COMPONENT	TASK	STATUS
Setup	Django 4+ & DRF Integration	COMPLETED
Database	PostgreSQL Schema & Docker Config	COMPLETED
Data Models	Customer & Loan Models (api/models.py)	COMPLETED
Ingestion	Background Worker Excel Import	COMPLETED
Logic	0-100 Credit Scoring Engine	COMPLETED
API: Register	POST /api/register	COMPLETED
API: Eligibility	POST /api/check-eligibility	COMPLETED
API: Create	POST /api/create-loan	COMPLETED
API: View Loan	GET /api/view-loan/<id>	COMPLETED
API: View Loans	GET /api/view-loans/<customer_id>	COMPLETED
Containerization	Full Docker Compose Orchestration	COMPLETED
Bonus	Unit Tests	NOT STARTED

2. TECHNICAL LOGIC DETAILS

- A. /register:
- Calculates 'approved_limit' = 36 * monthly_salary.
 - Rounds limit to the nearest lakh.
- B. /check-eligibility (The Scoring Engine):
- Weighs ratio of on-time payments.
 - Considers loan count and current year activity.
 - Rejection Criteria:
 - * Score automatically 0 if Debt > Approved Limit.
 - * Rejects if Score < 10.
 - * Rejects if New EMI + Current EMIs > 50% of monthly income.
 - Auto-Correction: Adjusts interest rates (Tiered: 12% or 16%) based on score

3. LOCAL SETUP PROCEDURE (Using Docker)

Follow these steps to run the project on your machine:

- Step 1: Build and Start Containers
- ```
docker-compose up --build
```
- Step 2: Initialize Database Tables
- ```
docker-compose exec web python manage.py migrate
```
- Step 3: Ingest Data from Excel
- ```
docker-compose exec web python manage.py ingest_data
```
- Step 4: Access Application
- Server URL: `http://localhost:8000`
- Admin URL: `http://localhost:8000/admin/`

4. PROJECT STACK

- Framework: Django 4.2.7, Django REST Framework
- Data: PostgreSQL 15, Redis 7 (Broker)
- Tasks: Celery 5.3.6
- Utils: Pandas, OpenPyXL (Excel Processing)
- Deployment: Docker Desktop

5. SUBMISSION GUIDELINES

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- Repository: Ensure all code is pushed to your GitHub.
  - Credentials: Default DB credentials provided in .env/docker-compose for easy review.

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Final Report Generated for Assignment Submission - Feb 2024